

# Overlap Does Not Decide Independence

Unit 2 · Topic 2.7 · THE PROBLEM

Suppose a school surveys 200 students about whether they own at least one pet and whether they play a musical instrument. Let  $P$  be the event that a randomly selected surveyed student owns a pet and  $I$  the event that the student plays an instrument.

**Aaliyah:** *The events overlap because 84 students do both. Since they are not mutually exclusive, they must be independent.*

**Marcus:** *Overlap does not decide independence; the relevant probabilities must be compared.*

## Pet ownership and instrument playing

| Pet?  | Plays | Does not | Total |
|-------|-------|----------|-------|
| Yes   | 84    | 36       | 120   |
| No    | 16    | 64       | 80    |
| Total | 100   | 100      | 200   |

### FIRST TAKE — one sentence before you work the parts

Does owning a pet seem to change the chance that a student plays an instrument? Give one table detail in one sentence.

- DOK 1** (a) State the counts for  $P$ ,  $I$ , and  $P \cap I$ .
- DOK 2** (b) Find  $P(I)$ ,  $P(I | P)$ , and  $P(P \cup I)$ . Show the fractions or setup.
- DOK 3** (c) Whose reasoning is supported? Decide independence using the two instrument-playing probabilities. Explain why the 84 students in both groups settle mutual exclusivity but do not settle independence. Write 3–4 sentences.

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.